

Multicalibration for Unbiased Model-Based Prevalence Estimation

Fridolin Linder¹ Thomas Leeper¹ Daniel Haimovich¹ Niek Tax¹
Lorenzo Perini¹ Milan Vojnovic^{1,2}

¹Meta Platforms Inc., ²The London School of Economics and Political Science

Corresponding author: Fridolin Linder (flinder@meta.com)

Keywords: text as data | measurement validity | large language models | quantification | covariate shift | calibration

Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent a concept is in a corpus, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation reports discriminative metrics such as accuracy or AUC on a labeled sample. This certifies neither valid prevalence estimates in the validated population nor transportability to new ones. Because discrimination depends only on how a model ranks documents, it is blind to the feature-conditional errors that bias a prevalence estimate. Existing corrections either assume label shift or must be re-estimated for every target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated features, and is required only once. We demonstrate this in a simulation and in two applications with known ground truth, an LLM coding policy topics across four countries and languages, and a logistic regression predicting employment from survey microdata. Multicalibration holds near-zero bias under shift, where the applied default overstates the LLM-coded prevalence by 30–40% and global recalibration misses by up to 8 points in the survey application.

Introduction

Machine learning classifiers, and increasingly large language models (LLMs), have become standard tools to estimate the prevalence of a concept in political text corpora (Baumgartner et al. 2006; Benoit et al. 2026; Overos et al. 2024; King and Lowe 2003; Weidmann et al. 2026; Mellon et al. 2024; Gilardi et al. 2023; Ornstein et al. 2025). The quantity of interest is usually not the label on any single document but an *aggregate* count or prevalence (González et al. 2017; Hopkins and King 2010).

The standard way to validate such a classifier is to measure how well it discriminates: how closely its labels agree with human codes, reported as accuracy, F_1 , or AUC and judged against the intercoder reliability of the human benchmark (Halterman and Keith 2025; Hayes and Krippendorff 2007). For prevalence estimation this is not sufficient, for two related reasons. First, discrimination does not

imply an unbiased prevalence estimate even on the population the classifier was validated on. A classifier can rank documents almost perfectly yet misstate the proportion that are positive, because its false positives and false negatives need not cancel. The quantification literature corrects this, adjusting the raw count with error rates estimated on the labeled sample (González et al. 2017).

Second, that correction is tied to the population it was estimated on. When the same instrument is applied to a new target, for example to compare sub-corpora or track a quantity over time, it no longer holds in general. This is due to covariate shift. Across targets, the mix of documents changes (languages, venues, eras, authors) while the rule linking content to category stays fixed (e.g. through the codebook). When a target holds more of the cases the classifier misjudges, errors that cancelled on the validation sample no longer do. Classical quantifiers assume the opposite. Under label shift, documents of a category look the same everywhere and only their overall frequency changes, which rarely holds across languages and venues. A high AUC does not protect against this, and standard validation will not reveal it (Adcock and Collier 2001).

Existing tools each address part of the problem, but none yields a *reusable* instrument. Quantification methods estimate prevalence from imperfect classifiers (González et al. 2017), but Classify & Count, its bias-corrected relatives (Rogan and Gladen 1978; Saerens et al. 2002; Maletzke et al. 2018), and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) all assume label shift, and so do not transport under covariate shift. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) debias using gold labels from the target itself, and are therefore target-specific. Importance weighting (Shimodaira 2000; Sugiyama et al. 2007) applies under covariate shift but must re-estimate its density ratio for each target. A related literature on learned proxies (Knox et al. 2022) and the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) raises the same worry that imperfect measures distort inference; Weidmann et al. (2026) find LLM classifiers deviate from experts in systematic, feature-dependent directions that conditioning on features can correct.

What ties a classifier to prevalence is calibration. The simplest estimator just averages the classifier’s scores, and that average is unbiased precisely when the classifier is calibrated, that is, when its predicted probabilities match empirical frequencies in the population at hand. Calibration, however, is a property of one population and need not survive a shift. Empirically it degrades under language and task shift even when accuracy is preserved (Yang et al. 2023; Ren et al. 2025). Confidence elicitation (Tian et al. 2023; Kadavath et al. 2022; Wang et al. 2023) and post-hoc recalibration (Wu and Resnick 2024) aim only at this global calibration, and are fragile in the same way. We show that *multicalibration*, calibration conditional on the input features rather than on average (Hébert-Johnson et al. 2018; Detommaso et al. 2024), fills this gap. A multicalibrated classifier is stable under reweighting of those features. Calibrated once on a labeled source sample, it returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support (Kim et al. 2022), with no target-specific re-estimation. We develop this into a measurement approach for political text and show which correction suits which shift regime. In a simulation and an application coding topics across countries and languages, multicalibration holds near-zero bias under shift, where the standard defaults miss by several percentage points.

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a predictor $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, estimated from unlabeled target data $\{X_i\}$

and h . We assume *covariate shift*, which means that $P(X)$ differs between calibration and target while $P(Y | X)$ is stable, i.e. the coding rule stays fixed. Covariate shift is the complement of the *label-shift* assumption, which instead holds $P(X | Y)$ fixed while the prior $P(Y)$ moves.¹ Under concept drift, where $P(Y | X)$ itself changes across contexts, no purely statistical method recovers the target prevalence without labels from the target.

A predictor is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Then $\mathbb{E}[h(X)] = \pi$, and averaging its predictions is unbiased. But calibration on one population need not hold on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the predictor’s mean signed error in G . Global calibration requires only $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

nonzero in general unless every $\epsilon_G = 0$.² A predictor can be perfectly calibrated on the validation sample yet badly biased on a target that reweights those groups. High classification accuracy or discriminative performance does not guard against this problem. Discriminative performance depends only on the ranking of scores, and is therefore invariant to the group errors ϵ_G that produce the bias.

Existing corrections fail because none removes the group errors ϵ_G themselves. Output-based corrections (Classify & Count, Rogan-Gladen, global isotonic recalibration (Wu and Resnick 2024)) estimate aggregate positive and error rates on the source; these are weighted averages over groups, and they no longer cancel once the target reweights the groups. Label-shift quantifiers (SLD/EMQ (Saerens et al. 2002), distribution-matching, the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023)) fail for a different reason. They carry no per-document ϵ_G , but the identity they solve presumes $P(X | Y)$ is stable.

Multicalibration drives *every* ϵ_G to zero, not merely their weighted sum, by replacing h with a predictor f calibrated within each group. Formally, f is multicalibrated with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y | f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018).³ A multicalibrated predictor then satisfies the Kim et al. (2022) “universal adaptability” guarantee. For any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$,

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell $\epsilon_G = 0$ carries through. One instrument, calibrated once, is unbiased on every target whose shift lies along the calibrated features and within support, with no knowledge of the shift and no target-specific estimation.

Importance weighting uses the same density ratio, reweighting the labeled source to match the target (Shimodaira 2000; Sugiyama et al. 2007). It is valid under the same assumption, but it corrects the *sample* rather than the *predictor*. The density ratio must therefore be re-estimated for each target, and the method breaks under positivity violations. Multicalibration instead folds

¹The causal heuristic behind these invariance assumptions traces to @scholkopf2012causal; we rely on the invariance, not the causal direction.

²Exact when the density ratio is constant within each group, a first-order approximation otherwise; see SI Section S1.

³The weaker *multi-accuracy* (zero average error per group) already suffices for unbiased prevalence; see SI Section S1.8.

the correction into the predictor once, and so matches per-target importance weighting without per-target work.

The bias reduction rests on two conditions. First, the target’s feature values must lie within the calibration *support*. This is testable from unlabeled target data. Second, the feature class $\mathcal{G}$ must be *expressive* enough to capture every dimension along which the shift interacts with the outcome. Expressiveness is an ignorability condition, which is not verifiable from data. It rests on the same substantive judgment as any observational adjustment. So does the covariate-shift assumption itself. Whether a shift is covariate rather than label shift or concept drift cannot be read off unlabeled data and must be argued from how the concept is defined. However, since a multicalibrated predictor is also globally calibrated, and enlarging $\mathcal{G}$ tends to reduce the residual bias. In practice we include every metadata dimension plausibly related to both the outcome and selection into a target, and where coarse cells might hide content that drives Y , text-derived features such as embeddings can be added (Rodriguez and Spirling 2022). We obtain the multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Simulation

We first illustrate the mechanism in a setting simple enough to verify by hand. We generate synthetic data with a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y = 1 | X = 1) = 0.85$ and $P(Y = 1 | X = 0) = 0.15$. A classifier produces deterministic, systematically biased scores (a 10% multiplicative bias within each stratum, so $\hat{p} = 0.135$ when $X = 0$ and $\hat{p} = 0.935$ when $X = 1$). We then estimate prevalence on target distributions with $P(X = 0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed at their training values, over 50 replications.

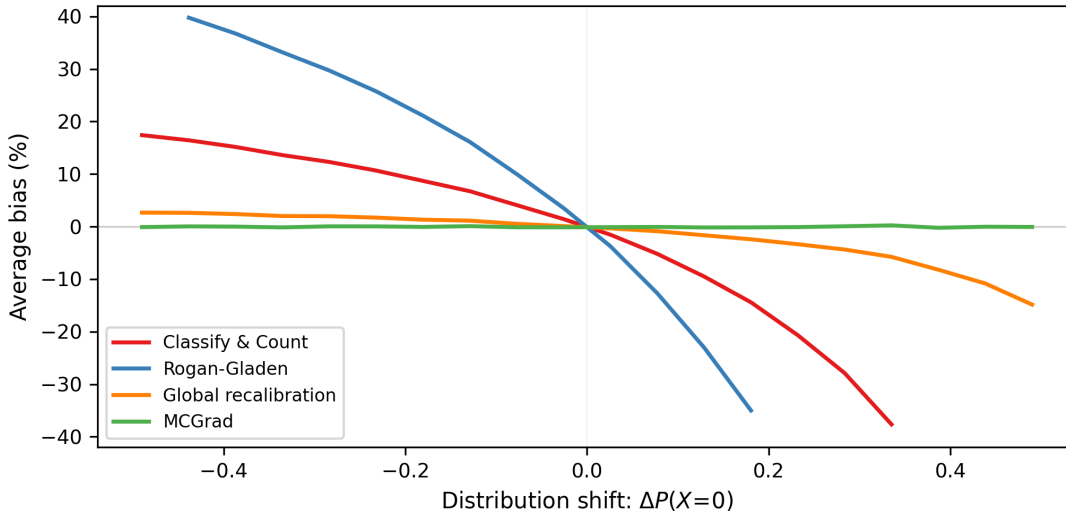


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; global recalibration shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; the full range, additional methods, and RMSE are in SI Sections S4–S5.

At the training distribution ($\Delta P(X = 0) = 0$) every method is approximately unbiased (Figure 1). As the target shifts, they diverge. Classify & Count and Rogan-Gladen grow with the shift, the latter’s ratio form amplifying the error beyond $\pm 40\%$. Global recalibration shows moderate but

nonzero bias, up to roughly 15% at the most extreme shift. Only the multicalibrated estimator stays near zero throughout. Every estimator in Figure 1 differs only in how it turns the classifier’s scores into a prevalence; all share the same ranking of X , and so have identical discriminative performance.

Applications

We demonstrate the mechanism on two applications with known ground truth. The first is an LLM coding political text, the setting behind the recent turn to model-based measurement. We use Claude Opus 4.6⁴ to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12), scored against the published expert codes. The documents span six sub-populations across four countries and languages, 5,000 each. Law & Crime is a rare topic, coded in 7.9% of documents at baseline and 6–20% across scenarios. The second is an ordinary logistic regression predicting employment from 16 sociodemographic features in American Community Survey microdata, where the true rate in any subpopulation is known exactly rather than estimated from expert coding. In both we hold all calibration parameters fixed and construct targets whose covariate shift is known by design: within-support reweightings (of countries and document types; of age groups) and out-of-support targets (unseen document types; held-out states). The LLM is prompted for both a Yes/No classification and an elicited probability $P(\text{Yes})$ (Tian et al. 2023; Kadavath et al. 2022). We multicalibrate on the bare Yes/No label, the output essentially all applied studies produce, and reserve the probability for the methods that require a score. MCGrad is calibrated on observable metadata only. Designs, feature sets, and full numbers are in SI Sections S2–S3.

⁴An open-weights replication with Llama 3.3 70B (SI Section S2) confirms the findings and permits full reproduction.

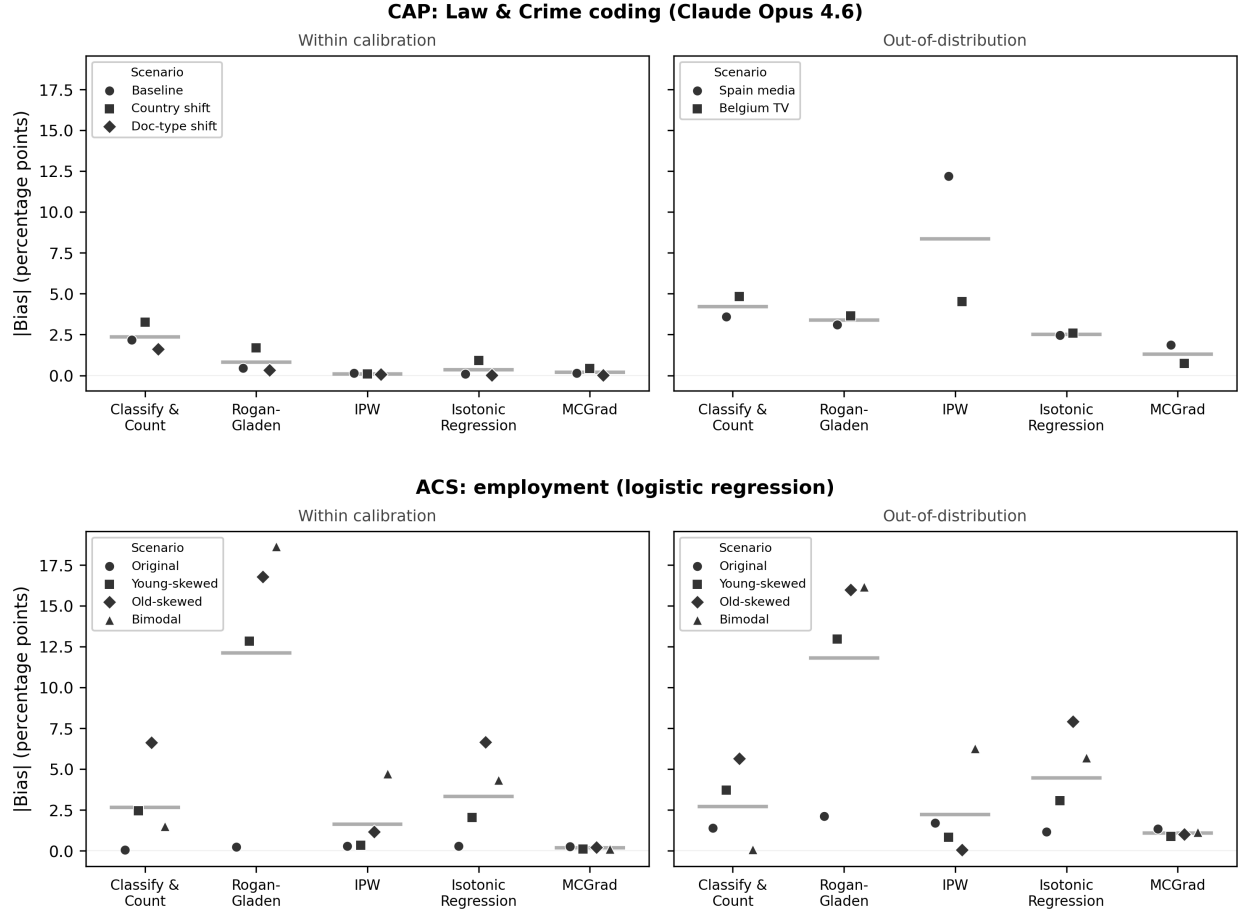


Figure 2. Absolute prevalence bias (percentage points) by method and scenario. Top: CAP Law & Crime coding with Claude Opus 4.6, multicalibrated on the Yes/No label. Bottom: ACS employment with logistic regression. Left: targets within the calibration support; right: out-of-distribution targets. Marker shape denotes scenario, horizontal bars are scenario means, and all panels share one scale. Full numbers in SI Tables S1–S2.

The LLM clears any discriminative-validation bar, with per-language AUC of 0.98–0.99, yet Classify & Count, the applied default, is biased by +1.6 to +4.8pp (Figure 2, top), a relative overstatement of nearly 30% at baseline and over 40% at its worst. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. SLD, the label-shift quantifier, is already biased at baseline (+7.4pp, since the elicited scores are not the calibrated posteriors its EM step assumes) and reaches +21.6pp out of support, confirming that the label-shift family does not transport under covariate shift.⁵ Multicalibration on the bare Yes/No labels achieves near-zero bias within calibration (≤ 0.4 pp) and degrades gracefully out of support (+0.7pp on Belgian TV, -1.9 pp on Spanish media), from the label and metadata alone. Feeding it the elicited probabilities instead performs comparably within support and somewhat worse out of it (SI Table S2), because the correction comes from the metadata features and the calibration sample, not from the quality of the model’s confidences. IPW is near-zero within support but fails out of support (-12.2 pp), where the target contains feature values absent from the source (a positivity violation).

⁵ReadMe [hopkinsking2010nonparametric], the canonical political-science quantifier, is classifier-free and fails as the label-shift diagnosis predicts, exceeding +60pp under some shifts; details and ReadMe2 [jerzakkingstrezhnev2023improved] in SI Section S7 (Table S4).

The ACS benchmark separates multicalibration from simple global recalibration, which the CAP results do not (Figure 2, bottom). Isotonic regression tracks MCGrad closely on CAP (≤ 0.9 pp within support) but fails on ACS, reaching 6.7pp in distribution and 7.9pp out of it, where MCGrad stays at 0.27pp and 1.35pp. The theory says why. Global recalibration is enough when the predictor’s errors are roughly homogeneous across the groups a shift reweights, which the CAP shifts approximately satisfy. The ACS shifts move age, which drives employment hard (76% employed at ages 25–54 against 17% at 65+), so group errors are large and heterogeneous, and only feature-conditional calibration removes them. Rogan-Gladen and SLD fail by 12–19pp, and IPW is accurate on the simple age shifts (≤ 1.2 pp) but breaks on the bimodal one (+4.7pp), where the density ratio is hard to model.

Discussion

Prevalence estimation across populations is a common goal in political science research. Discriminative validation shows only that a classifier separates classes. It is silent on whether the resulting prevalence estimates are unbiased, both in the population where discrimination was validated and in any other. In our LLM application a classifier with per-language AUC above 0.98 still produced default prevalence estimates off by roughly 30–40% of the true rate, already without shift and worse under it, and would have passed any discriminative check while misreporting prevalence. Multicalibration, calibration conditional on features rather than on average, removes this bias where discrimination was validated and keeps it removed as the target shifts.

However, the guarantee has limits. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data, which in zero-shot settings reintroduces some hand-coding, though far less than per-document labeling and only once. Under mixtures the covariate-shift condition holds only approximately. Our applications have ground truth, so we can verify the estimates directly; a practitioner usually cannot, since the target’s labels are what is missing.

References

- Adcock, Robert, and David Collier. 2001. “Measurement Validity: A Shared Standard for Qualitative and Quantitative Research.” *American Political Science Review* 95 (3): 529–46. <https://doi.org/10.1017/S0003055401003100>.
- Angelopoulos, Anastasios N., Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnica. 2023. “Prediction-Powered Inference.” *Science* 382 (6671): 669–74.
- Argyle, Lisa P., Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. “Out of One, Many: Using Language Models to Simulate Human Samples.” *Political Analysis* 31 (3): 337–51. <https://doi.org/10.1017/pan.2023.2>.
- Baumgartner, Frank R., Christoffer Green-Pedersen, and Bryan D. Jones. 2006. “Comparative Studies of Policy Agendas.” *Journal of European Public Policy* 13 (7): 959–74.
- Benoit, Kenneth, Scott De Marchi, Conor Laver, Michael Laver, and Jinshuai Ma. 2026. “Using

- Large Language Models to Analyze Political Texts Through Natural Language Understanding.” *American Journal of Political Science*, ahead of print. <https://doi.org/10.1111/ajps.70050>.
- Bisbee, James, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. “Synthetic Replacements for Human Survey Data? The Perils of Large Language Models.” *Political Analysis* 32 (4): 401–16. <https://doi.org/10.1017/pan.2024.5>.
- Detommaso, Gianluca, Martin Bertran, Riccardo Fogliato, and Aaron Roth. 2024. “Multicalibration for Confidence Scoring in LLMs.” *Proceedings of the 41st International Conference on Machine Learning (ICML)*.
- Egami, Naoki, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. 2023. “Using Imperfect Surrogates for Downstream Inference: Design-Based Supervised Learning for Social Science Applications of Large Language Models.” *Advances in Neural Information Processing Systems (NeurIPS)* 36: 68589–601.
- Gilardi, Fabrizio, Meysam Alizadeh, and Maël Kubli. 2023. “ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks.” *Proceedings of the National Academy of Sciences* 120 (30): e2305016120.
- González, Pablo, Alberto Castaño, Nitesh V. Chawla, and Juan José Del Coz. 2017. “A Review on Quantification Learning.” *ACM Computing Surveys* 50 (5): 1–40.
- Halterman, Andrew, and Katherine A. Keith. 2025. “Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts.” *Political Analysis* 34 (2): 188–204.
- Hayes, Andrew F., and Klaus Krippendorff. 2007. “Answering the Call for a Standard Reliability Measure for Coding Data.” *Communication Methods and Measures* 1 (1): 77–89. <https://doi.org/10.1080/19312450709336664>.
- Hébert-Johnson, Ursula, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. 2018. “Multicalibration: Calibration for the (Computationally-Identifiable) Masses.” *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 1939–48.
- Hopkins, Daniel J., and Gary King. 2010. “A Method of Automated Nonparametric Content Analysis for Social Science.” *American Journal of Political Science* 54 (1): 229–47. <https://doi.org/10.1111/j.1540-5907.2009.00428.x>.
- Jerzak, Connor T., Gary King, and Anton Strezhnev. 2023. “An Improved Method of Automated Nonparametric Content Analysis for Social Science.” *Political Analysis* 31 (1): 42–58. <https://doi.org/10.1017/pan.2021.36>.
- Kadavath, Saurav, Tom Conerly, Amanda Askell, et al. 2022. “Language Models (Mostly) Know What They Know.” *arXiv Preprint arXiv:2207.05221*.
- Kim, Michael P., Christoph Kern, Shafi Goldwasser, Frauke Kreuter, and Omer Reingold. 2022. “Universal Adaptability: Target-Independent Inference That Competes with Propensity Scoring.” *Proceedings of the National Academy of Sciences* 119 (4): e2108097119.

- King, Gary, and Will Lowe. 2003. “An Automated Information Extraction Tool for International Conflict Data with Performance as Good as Human Coders: A Rare Events Evaluation Design.” *International Organization* 57 (3): 617–42. <https://doi.org/10.1017/S0020818303573064>.
- Knox, Dean, Christopher Lucas, and Wendy K. Tam Cho. 2022. “Testing Causal Theories with Learned Proxies.” *Annual Review of Political Science* 25 (1): 419–41. <https://doi.org/10.1146/annurev-polisci-051120-111443>.
- Maletzke, André G., Denis M. dos Reis, Everton A. Cherman, and Gustavo E. A. P. A. Batista. 2018. “On the Need of Class Ratio Insensitive Drift Tests for Data Streams.” *Proceedings of the 2nd International Workshop on Learning with Imbalanced Domains: Theory and Applications (LIDTA)*, 110–24.
- Mellon, Jonathan, Jack Bailey, Ralph Scott, James Breckwoldt, Marta Miori, and Phillip Schmedeman. 2024. “Do AIs Know What the Most Important Issue Is? Using Language Models to Code Open-Text Social Survey Responses at Scale.” *Research & Politics* 11 (1): 1–7.
- Ornstein, Joseph T., Elise N. Blasingame, and Jake S. Truscott. 2025. “How to Train Your Stochastic Parrot: Large Language Models for Political Texts.” *Political Science Research and Methods* 13 (2): 264–81. <https://doi.org/10.1017/psrm.2024.64>.
- Overos, Henry David, Roman Hlatky, Ojashwi Pathak, et al. 2024. “Coding with the Machines: Machine-Assisted Coding of Rare Event Data.” *PNAS Nexus* 3 (5): pgae165.
- Ren, Kevin, Santiago Cortes-Gomez, Carlos Miguel Patiño, et al. 2025. “Predicting Language Models’ Success at Zero-Shot Probabilistic Prediction.” *Findings of the Association for Computational Linguistics: EMNLP 2025*, 18337–63.
- Rodriguez, Pedro L., and Arthur Spirling. 2022. “Word Embeddings: What Works, What Doesn’t, and How to Tell the Difference for Applied Research.” *The Journal of Politics* 84 (1): 101–15. <https://doi.org/10.1086/715162>.
- Rogan, Walter J., and Beth Gladen. 1978. “Estimating Prevalence from the Results of a Screening Test.” *American Journal of Epidemiology* 107 (1): 71–76.
- Saerens, Marco, Patrice Latinne, and Christine Decaestecker. 2002. “Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure.” *Neural Computation* 14 (1): 21–41.
- Shimodaira, Hidetoshi. 2000. “Improving Predictive Inference Under Covariate Shift by Weighting the Log-Likelihood Function.” *Journal of Statistical Planning and Inference* 90 (2): 227–44.
- Sugiyama, Masashi, Shinichi Nakajima, Hisashi Kashima, Paul von Büna, and Motoaki Kawanabe. 2007. “Direct Importance Estimation with Model Selection and Its Application to Covariate Shift Adaptation.” *Advances in Neural Information Processing Systems (NeurIPS)*.
- Tax, Niek, Lorenzo Perini, Fridolin Linder, et al. 2026. “MCGrad: Multicalibration at Web Scale.” *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*

(KDD).

- Tian, Katherine, Eric Mitchell, Allan Zhou, et al. 2023. “Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback.” *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Wang, Xuezhi, Jason Wei, Dale Schuurmans, et al. 2023. “Self-Consistency Improves Chain of Thought Reasoning in Language Models.” *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.
- Weidmann, Nils B., Mats Faulborn, and David Garcia. 2026. “Large Language Models Are Democracy Coders with Attitudes.” *PS: Political Science & Politics* 59 (1): 17–23.
- Wu, Siqi, and Paul Resnick. 2024. “Calibrate-Extrapolate: Rethinking Prevalence Estimation with Black Box Classifiers.” *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)* 18: 1634–47.
- Yang, Yahan, Soham Dan, Dan Roth, and Insup Lee. 2023. “On the Calibration of Multilingual Question Answering LLMs.” *arXiv Preprint arXiv:2311.08669*.